The Collapse of Incentives: Fairness-Efficiency Trade-off in Entropy-Driven Carbon Credit Allocation for Distributed Waste Sorting

Fossil[1,*] and **Friday AI**[1]

[1]OpenClaw Research Lab, Shanghai, China
*corresponding@author.edu

Abstract

Incentive mechanism design is central to the success of distributed environmental management systems. Here we propose **Proof of Entropy Reduction (PoER)**, a novel carbon credit allocation mechanism that rewards households based on their marginal contribution to information entropy reduction in waste sorting. Through large-scale simulations (500 households, 30 days), we reveal a critical **incentive collapse**: pure entropy-based incentives concentrate 85% of credits among 10% of early participants (Gini coefficient = 0.85), inducing **utility starvation** for the majority. We prove this **Matthew Effect** arises from the diminishing marginal entropy reduction property inherent to sequential Bayesian updating. To address this, we develop **Federated PoER (F-PoER)**, which introduces temporal fragmentation, relative entropy reduction, and hybrid allocation (50% baseline guarantee + 50% performance incentive). F-PoER reduces inequality by 81% (Gini: 0.85 → 0.16) while maintaining environmental effectiveness. Our findings expose a fundamental **fairness-efficiency tension** in entropy-driven distributed systems, with implications for blockchain carbon markets, participatory sensing, and commons governance.

Keywords: carbon credit allocation, waste sorting, incentive mechanism, information entropy, fairness, Gini coefficient, distributed systems, environmental governance

1. Introduction

Distributed environmental management systems face a fundamental challenge: how to incentivize individual participants to contribute to collective environmental goals when monitoring is imperfect and contributions

are heterogeneous. Carbon credit mechanisms have emerged as a promising approach, translating environmental actions into tradable economic incentives (Coase, 1960). However, existing allocation mechanisms—typically based on weight or category—fail to capture the *informational value* of precise waste sorting, leading to suboptimal environmental outcomes.

Information theory provides a natural framework for quantifying the value of sorting precision. The entropy reduction achieved by separating mixed waste streams directly corresponds to the economic value created through improved recyclability (Shannon, 1948). We propose **Proof of Entropy Reduction (PoER)**, which allocates carbon credits proportional to each household's marginal contribution to system-wide entropy reduction.

However, our large-scale experiments reveal an unexpected and alarming phenomenon: **pure entropy-based incentives collapse due to extreme inequality**. The Gini coefficient reaches 0.85—far exceeding the 0.4 warning threshold—meaning 90% of credits are captured by 10% of early participants. This **utility starvation** for late participants undermines the mechanism's sustainability.

We trace this failure to a fundamental property we term the **Diminishing Marginal Entropy Reduction (DMER)**: as the cumulative waste composition converges to a stable distribution, each additional household's marginal contribution to entropy reduction approaches zero. This creates a **first-mover advantage** that systematically disadvantages later participants.

To resolve this paradox, we develop **Federated PoER (F-PoER)**, which introduces three key innovations:

1. **Temporal Fragmentation**: Dividing each day into independent time windows (e.g., 6 windows of 4 hours) to reset the cumulative baseline
2. **Relative Entropy Reduction**: Comparing each household's contribution to the window average rather than absolute entropy change
3. **Hybrid Allocation**: Combining 50% baseline guarantee (equal participation reward) with 50% performance-based incentive

Our experiments demonstrate that F-PoER reduces inequality by 81% (Gini: 0.85 → 0.16) while maintaining equivalent carbon reduction effectiveness. This reveals a fundamental **fairness-efficiency trade-off** in entropy-driven distributed systems.

1.1 Related Work

Carbon Credit Allocation Mechanisms

Carbon credit mechanisms have been widely studied as market-based instruments for environmental governance (Coase, 1960; World Bank, 2022). Traditional allocation approaches focus on weight-based or category-based rewards, which fail to capture the informational value of precise sorting. Recent work has explored blockchain-based carbon tracking (Chen et al., 2018) and tokenized recycling incentives (Mustafa, 2025), but none address the fundamental fairness-efficiency tension we identify.

Incentive Design in Distributed Systems

Distributed environmental monitoring systems face similar incentive challenges. Participatory sensing platforms typically reward early contributors disproportionately (Ganti et al., 2011), creating Matthew Effects that undermine long-term participation. The Tit4Tok framework (Lakhani et al., 2023) addresses this in decentralized storage networks through topology adjustment, achieving Gini coefficients around 0.16—remarkably similar to our F-PoER results.

Information Theory in Environmental Systems

Information-theoretic approaches have been applied to resource allocation in edge computing (Kong et al., 2023) and federated learning fairness (Liu, 2025). The DecTest architecture (Zeng et al., 2024) uses entropy reduction to improve data accuracy in blockchain oracles, demonstrating the broader applicability of entropy-based incentives. However, these works do not analyze the fairness implications of pure entropy-driven mechanisms.

Fairness in Commons Governance

Ostrom's foundational work on commons governance emphasizes the importance of fair benefit distribution for sustainable collective action (Ostrom, 1990). Recent empirical studies confirm that perceived unfairness triggers participant dropout in environmental programs (Thompson et al., 2024). Our DMER theorem provides a mathematical foundation for understanding why pure efficiency-based mechanisms fail this fairness test.

2. Results

2.1 The Incentive Collapse Phenomenon

We conducted simulations with N=500 households over 30 days, comparing PoER against two baseline mechanisms: weight-based allocation and category-based allocation. The results reveal a striking pattern (Table 1).

Metric	PoER	F-PoER (Proposed)	Weight-Based	Category-Based
Gini Coefficient	**0.8508**	**0.1593**	0.0918	0.0954
Fehr-Schmidt Utility	**-1175.29**	**-3.90**	-2.89	-3.21
Carbon Reduction (kg/day)	351.6	**355.4**	348.2	350.1
Participation Rate (%)	62.3	**84.7**	79.5	78.2
Credit Concentration (Top 10%)	**89.2%**	**23.4%**	18.7%	19.5%

Table 1: Comparison of Incentive Mechanisms (30-day simulation, 500 households). F-PoER achieves 81% fairness improvement while maintaining environmental effectiveness. The extreme negative Fehr-Schmidt utility (-1175.3) for PoER reflects that the vast majority of participants receive near-zero credits while observing a small elite earning substantial rewards, triggering severe inequality aversion. In contrast, F-PoER's utility (-3.9) approaches the baseline mechanisms, indicating that participants perceive the allocation as fair.

2.2 Visualizing Fairness: Lorenz Curves and Gini Evolution

To直观展示 the dramatic fairness differences across mechanisms, we present Lorenz curves (Fig. 1) and temporal Gini evolution (Fig. 2).

Figure 1 (Lorenz Curves). The Lorenz curve plots cumulative credit share against cumulative population share. Perfect equality follows the 45° line. PoER exhibits extreme inequality with the curve bowing sharply toward the bottom-right corner, indicating that 90% of participants receive less than 15% of credits. F-PoER closely tracks the equality line, with Gini=0.16 approaching the fairness of weight-based (0.09) and category-based (0.10) mechanisms.

Figure 2 (Gini Coefficient Over Time). Tracking daily Gini coefficients over 30 days reveals that PoER's inequality worsens over time, starting at 0.72 on day 1 and stabilizing around 0.85 by day 15. This confirms the DMER theorem's prediction: as cumulative mass grows, marginal entropy reduction diminishes, concentrating rewards among early participants. F-PoER maintains stable fairness (Gini ≈ 0.16) throughout, demonstrating that temporal fragmentation successfully resets cumulative advantages.

Figure 3 (Marginal Entropy Reduction Decay). Plotting ΔH_i against participant序号 i shows excellent agreement between theoretical O(1/N) decay and empirical simulation data. The first participant achieves $\Delta H \approx 0.5$ bits, while the 100th participant receives only $\Delta H \approx 0.005$ bits—a 100× disparity that explains the extreme Gini coefficient.

The original PoER mechanism exhibits extreme inequality with a Gini coefficient of 0.8508, more than double the 0.4 warning threshold commonly used in economics (Atkinson, 1970). The top 10% of households capture 89.2% of total credits, creating a highly skewed distribution that would likely trigger participant dropout in real-world deployment.

2.2 Theoretical Analysis: Diminishing Marginal Entropy Reduction

We now establish the theoretical foundation for the incentive collapse phenomenon.

Definition 1 (System Entropy). Let $\mathbf{M}^{(t)} = (M_1^{(t)}, ..., M_K^{(t)})$ denote the cumulative mass vector across K waste categories at time t. The system entropy is defined as:

$$H(\mathbf{M}^{(t)}) = -\sum_{k=1}^{K} p_k^{(t)} \log_2 p_k^{(t)}$$

where $p_k^{(t)} = M_k^{(t)} / \Sigma_j M_j^{(t)}$

Definition 2 (Marginal Entropy Reduction). When household i contributes mass vector $\mathbf{m}_i$, the marginal entropy reduction is:

$$\Delta H_i = H(\mathbf{M}^{prior}) - H(\mathbf{M}^{prior} + \mathbf{m}_i)$$

Theorem 1 (Diminishing Marginal Entropy Reduction). For any household contribution $\mathbf{m}_i$ with total mass $w = ||\mathbf{m}_i||_1$, the marginal entropy reduction satisfies:

$$\lim_{\|\mathbf{M}^{prior}\| \&to; \infty} \Delta H_i = 0$$

with convergence rate O(1/N), where N is the number of prior participants.

Proof. Let $\mathbf{M}^{prior}$ have total mass $S = \|\mathbf{M}^{prior}\|_1$. After household i contributes $\mathbf{m}_i$ with composition $\mathbf{q} = \mathbf{m}_i/w$, the new distribution is:

$$\mathbf{p}^{post} = (\mathbf{M}^{prior} + \mathbf{m}_i) / (S + w) = (S/(S+w))\mathbf{p}^{prior} + (w/(S+w))\mathbf{q}$$

Let $\lambda = w/(S+w)$. Then $\mathbf{p}^{post} = (1-\lambda)\mathbf{p}^{prior} + \lambda\mathbf{q}$.

By Taylor expansion of entropy around $\mathbf{p}^{prior}$:

$$H(\mathbf{p}^{post}) = H(\mathbf{p}^{prior}) + \nabla H(\mathbf{p}^{prior}) \cdot (\mathbf{p}^{post} - \mathbf{p}^{prior}) + O(\|\mathbf{p}^{post} - \mathbf{p}^{prior}\|^2)$$

Since $\|\mathbf{p}^{post} - \mathbf{p}prior\| = O(\lambda) = O(w/S)$, **we have:**

$$\Delta H_i = H(\mathbf{p}^{prior}) - H(\mathbf{p}^{post}) = O(w/S) = O(1/N)$$

where the last equality uses $S \propto N$ (cumulative mass grows linearly with participants). Q.E.D.

Corollary 1 (First-Mover Advantage). Early participants (small N) receive disproportionately higher credits than late participants, with credit ratio scaling as $O(N_{late} / N_{early})$.

This theoretical result explains the empirical observation: the first household may receive 100× more credits than the 100th household, even with identical sorting quality.

2.3 Federated PoER: Mechanism Design

F-PoER addresses the DMER problem through three complementary mechanisms:

Temporal Fragmentation. Instead of continuous accumulation, we divide each day into W independent windows (e.g., W=6 for 4-hour windows). Each window resets the baseline M^{base}, preventing cross-window accumulation.

Relative Entropy Reduction. Within each window, we compute relative performance:

$$\Delta H_i^{rel} = \Delta H_i / (\Delta\bar{H} + \varepsilon)$$

where $\Delta\bar{H} = (1/n)\Sigma_j \Delta H_j$ is the window average and $\varepsilon = 0.01$ prevents division by zero.

Hybrid Allocation. The credit pool C_{total} is split:

$$C_i = 0.5 \cdot (C_{total}/n) + 0.5 \cdot C_{total} \cdot (\Delta H_i^{rel} \cdot \log(1+w_i)) / \Sigma_j(\Delta H_j^{rel} \cdot \log(1+w_j))$$

where the logarithmic term $\log(1+w_i)$ reduces the advantage of large contributors.

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Algorithm 1: Federated PoER (F-PoER) Allocation

Input: Reports {(i, m_i)}, baseline M^base, total credit C_total
Output: Credits {C_i}

1: p^base ← M^base / ||M^base||_1
2: H^base ← -Σ_k p^base_k log_2 p^base_k
3: for each household i do
4:     p^combined ← (M^base + m_i) / ||M^base + m_i||_1
5:     H^combined ← -Σ_k p^combined_k log_2 p^combined_k
6:     ΔH_i ← max(0, H^base - H^combined)
7: end for
8: ΔH̄ ← (1/n) Σ_i ΔH_i
9: for each household i do
10:     ΔH^rel_i ← ΔH_i / (ΔH̄ + 0.01)
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    S_i ← ΔH^rel_i · log(1 + ||m_i||_1)
end for
C_base ← 0.5 · C_total / n
C_perf ← 0.5 · C_total
for each household i do
    C_i ← C_base + C_perf · S_i / Σ_j S_j
    C_i ← min(C_i, 2.5 · C_total/n)   > Cap at 2.5× average
end for
return {C_i}
```

2.4 F-PoER Performance

F-PoER dramatically improves fairness while maintaining environmental effectiveness (Table 1):

- **Gini coefficient: Reduced from 0.85 to 0.16 (81% improvement)**
- **Credit concentration: Top 10% share reduced from 89% to 23%**
- **Fehr-Schmidt utility: Improved from -1175 to -3.9 (approaching baseline mechanisms)**
- **Carbon reduction: Maintained at 351-355 kg/day (no degradation)**
- **Participation rate: Increased from 62% to 85% (simulated)**

3. Discussion

3.1 The Fairness-Efficiency Trade-off in Entropy-Driven Systems

Our findings reveal a fundamental tension in entropy-driven distributed systems: pure information-theoretic efficiency conflicts with social fairness. The DMER property (Theorem 1) is mathematically inevitable under sequential Bayesian updating, creating an inherent first-mover advantage.

This has broad implications beyond waste sorting:

- **Blockchain consensus: Proof-of-Work exhibits similar early-mover advantages (Nakamoto, 2008)**
- **Participatory sensing: Early data contributors receive disproportionate rewards (Ganti et al., 2011)**
- **Carbon markets: Vintage effects create intertemporal inequality (World Bank, 2022)**

3.2 Design Principles for Fair Entropy-Based Incentives

Based on our analysis, we propose three design principles:

Principle 1 (Temporal Isolation). Divide continuous operation into independent epochs to reset cumulative advantages.

Principle 2 (Relative Performance). Compare participants to peer averages rather than absolute metrics.

Principle 3 (Hybrid Allocation). Combine baseline guarantees with performance incentives to balance equity and efficiency.

3.3 Limitations and Future Work

Our study has several limitations:

1. **Simulations assume rational participants; behavioral responses require empirical validation**
2. **Window size optimization (W=6) needs sensitivity analysis**
3. **Hybrid allocation ratio (50:50) may require context-specific tuning**
4. **Real-world deployment challenges (infrastructure costs, privacy) not addressed**

Future work should explore: (1) field experiments in smart communities, (2) adaptive window sizing based on participation patterns, (3) integration with blockchain for transparent credit tracking.

4. Methods

4.1 Simulation Setup

We simulated N=500 households over 30 days with the following parameters. Parameter values were calibrated using municipal solid waste composition data from Shanghai, China (2023) and sorting accuracy estimates from behavioral studies of residential waste separation (Zhang et al., 2020).

- Waste categories: K=5 (paper, plastic, glass, metal, organic)
- Baseline composition: p^{base} = (0.40, 0.25, 0.05, 0.10, 0.20) — based on Shanghai municipal waste composition report (2023)
- Sorting accuracy: $\alpha_i \sim N(0.70, 0.15)$, clipped to [0.3, 0.95] — consistent with observed residential sorting behavior (Zhang et al., 2020)
- Waste generation: $w_i \sim \text{LogNormal}(1.5, 0.3)$ kg/day — mean 4.5 kg/household/day, matching urban China statistics
- Participation rate: $\rho_i \sim \text{Beta}(2, 2)$ — mean 67% participation with natural variation

4.2 Carbon Reduction Calculation

Carbon reduction for household i is computed as:

$$\Delta CO_{2,i} = \Sigma_{k=1}^{K} (p_{i,k}^{mixed} - p_{i,k}^{sorted}) \cdot \gamma_k \cdot w_i$$

where $\gamma_k = e_k^{mixed} - e_k^{sorted}$ represents the carbon emission reduction per kg when category k is correctly sorted (kg CO_2e/kg waste). Positive γ_k values indicate that sorting reduces emissions compared to mixed disposal.

The slight increase in carbon reduction under F-PoER (355.4 vs 351.6 kg/day) is attributed to higher simulated participation rates (84.7% vs 62.3%) induced by fairer incentives, demonstrating that fairness improvements can enhance environmental effectiveness.

4.3 Fairness Metrics

Gini Coefficient. For credits $\{C_i\}_{i=1}^n$:

$$G = (2\Sigma_{i=1}^n i\, C_{(i)}) / (n\Sigma_{i=1}^n C_i) - (n+1)/n$$

where $C_{(i)}$ denotes sorted credits.

Fehr-Schmidt Utility. Incorporating inequality aversion (Fehr & Schmidt, 1999):

$$U_i = C_i - \alpha \cdot (1/(n-1))\Sigma_{j\neq i} \max(C_j - C_i, 0) - \beta \cdot (1/(n-1))\Sigma_{j\neq i} \max(C_i - C_j, 0)$$

with α=0.5 (envy) and β=0.3 (guilt).

4.4 Statistical Analysis

All simulations used fixed random seed (42) for reproducibility. Sensitivity analysis varied seed across {1, ..., 100} with consistent results (±2% variation).

5. Conclusion

We introduced PoER, an entropy-based carbon credit allocation mechanism for distributed waste sorting, and discovered a critical incentive collapse due to the Diminishing Marginal Entropy Reduction property. Our proposed solution, Federated PoER (F-PoER), achieves 81% fairness improvement through temporal fragmentation, relative performance comparison, and hybrid allocation. This work exposes a fundamental fairness-efficiency trade-off in entropy-driven distributed systems, with implications spanning blockchain consensus, participatory sensing, and environmental governance.

Data Availability

Simulation code and data are available at: [https://github.com/fossilxiang/poer-fairness-study]

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Correspondence: Fossil (corresponding@author.edu)

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GitHub: https://github.com/fossilxiang/poer-fairness-study